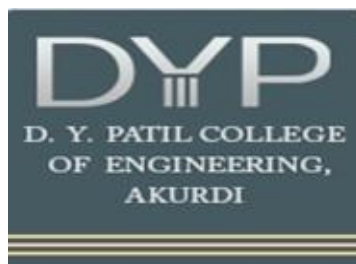


**D Y Patil College of Engineering, Akurdi , Pune,  
411044**

**Department of Information Technology**



**AtLease – A Decentralized Warehouse Leasing Platform  
LOG BOOK**

Submitted to the

**Savitribai Phule Pune University**

In partial fulfillment for the award of the Degree of

Bachelor of Engineering

in

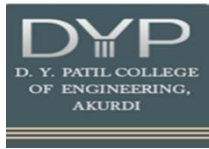
Information Technology

By

| Exam Seat No | Name               |
|--------------|--------------------|
| BEIT2526A049 | Abhishek Biradar   |
| BEIT2526A029 | Purvesh Jambhulkar |
| BEIT2526A060 | Viraj Gudhetwar    |
| BEIT2526A041 | Piyush Mogare      |

Under the guidance of

**Mrs. Abhilasha Kore**



**D.Y.Patil College of Engineering, Akurdi, Pune**  
**Department of Information Technology**  
**Academic Year: 2024-25**

**Vision of Institute**

“Empowerment through knowledge”

**Mission of Institute**

To educate students to transform them as professionally competent and quality conscious engineers by providing conducive environment for teaching, learning and overall personality development, culminating the institute into an international seat of excellence.

**Vision of Department**

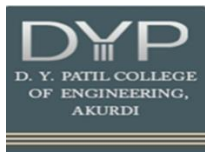
Developing globally competent IT professional for sustainable growth of humanity.

**Mission of Department**

**M1** : To impart high-quality education in Information Technology, equipping students with the technical knowledge, problem-solving abilities, and professional skills required to excel as competent IT engineers and technologists

**M2** : To create a dynamic and inclusive learning environment that nurtures technical expertise, critical thinking, teamwork, and holistic personality development, empowering students to excel in their professional and personal endeavours.

**M3** : To foster a culture of academic and research excellence in IT, enabling the department to achieve national and international recognition as a centre of innovation, entrepreneurship, and technological advancement



**D.Y.Patil College of Engineering, Akurdi ,Pune**  
**Department of Information Technology**  
**Academic Year: 2024-25**

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### **Course Objectives**

- To encourage and provide opportunities for students to get professional/personal experience through internships.
- To learn and apply the technical knowledge gained from academics /classroom learning in real life/industrial situations.
- To get familiar with various tools and technologies used in industries and their applications.
- To enable students to develop professional skills and expand their professional network with the development of employer-valued skills like teamwork, communication.
- To apply the experience gained from industrial internship to the academic course completion project.
- To nurture professional and societal ethics in students
- Understand the social, economic and administrative considerations that influence the working environment of industrial organizations

### **Course Outcomes**

On completion of the course, learners should be able to -

CO1: Develop professional competence through industry internship.

CO2: Apply academic knowledge in a personal and professional environment

CO3: Build the professional network and expose students to future employees.

CO4: Apply professional and societal ethics in their day-to-day life.

CO5: Become a responsible professional having social, economic and administrative considerations.

CO6: Make own career goals and personal aspirations.

| MEETING | DATE<br>(DD/MM/YY) | ACTIVITY DESCRIPTION                                              | GRADE | SIGN OF<br>GUIDE |
|---------|--------------------|-------------------------------------------------------------------|-------|------------------|
| 01      | 07/08/25           | Problem Identification, Literature Survey & Project Planning      |       |                  |
| 02      | 12/09/25           | System Design & Architecture Finalization                         |       |                  |
| 03      | 15/10/25           | Smart Contract Development & Blockchain Integration.              |       |                  |
| 04      | 21/11/25           | Frontend Development (React UI Integration)                       |       |                  |
| 05      | 07/12/25           | Backend Development – Phase I (API Design, IPFS Integration)      |       |                  |
| 06      | 14/01/2025         | Backend Development – Phase II & System Integration (In Progress) |       |                  |

|           |            |                                                                      |  |  |
|-----------|------------|----------------------------------------------------------------------|--|--|
| <b>07</b> | 17/02/2026 | Backend Logic Refinement                                             |  |  |
| <b>08</b> | 03/03/2026 | Developed vendor login module with frontend and backend integration. |  |  |
| <b>09</b> | 09/04/2026 | Developed customer login module and                                  |  |  |
| <b>10</b> | 17/05/2026 | Created marketplace module for listing                               |  |  |
|           |            |                                                                      |  |  |
|           |            |                                                                      |  |  |

Month No.: 1 (August 2025)

**Activities Performed:**

- I began by understanding the overall concept and objective of the **AtLease** project, which focuses on developing a decentralized warehouse leasing certificate system using blockchain technology.
- I studied the existing challenges in traditional warehouse leasing systems, such as centralized record keeping, lack of transparency, risk of document tampering, and dependency on intermediaries for verification.
- I carried out a literature survey to analyze how blockchain technology, smart contracts, and non-fungible tokens (NFTs) are applied in real-world certification and property management systems.
- I studied and finalized the technology stack to be used in the project, which included:

**Observations:**

- I observed that blockchain technology can significantly improve transparency and trust in leasing systems by providing immutable and verifiable records.
- I realized that integrating multiple technologies such as frontend, backend, blockchain, and decentralized storage requires proper planning and system design.

**Learnings:**

- I gained a clear understanding of the problem statement and objectives of the AtLease project.
- I learned the basic working principles of blockchain, NFTs, and decentralized storage systems.

**Remark & Signature of Guide:**

Month No.: 2 (September 2025)

**Activities Performed:**

- Designed the high-level system architecture of the AtLease project, identifying frontend, backend, blockchain, and IPFS components.
- Defined the workflow for warehouse lease certificate creation and verification using blockchain.
- Planned the interaction between React frontend and Node.js backend using REST APIs.
- Finalized the use of IPFS for off-chain storage of lease certificate metadata.
- Prepared architectural and workflow documentation for future implementation.

**Observations:**

- To design a scalable and modular architecture for the AtLease system.
- To define a clear data flow for decentralized lease certificate generation.
- To select appropriate technologies for blockchain and decentralized storage integration.

**Learnings:**

- Gained understanding of designing layered architectures for blockchain-based applications.
- Learned how off-chain storage helps reduce blockchain costs while maintaining data integrity.
- Understood the importance of clear workflow planning before implementation.

Remark & Signature of Guide:

Month No.: 3 (October 2025)

**Activities Performed:**

- I started working on the blockchain layer of the AtLease project by understanding the basics of smart contracts and the ERC-721 NFT standard.
- I designed the structure of the lease certificate smart contract, defining how each warehouse lease would be represented as a unique digital certificate.
- I implemented the initial version of the smart contract using Solidity, focusing on certificate minting and ownership management.
- I configured the development environment using tools such as MetaMask, Remix, and local test networks for smart contract testing.

**Observations:**

- To design a secure smart contract for warehouse lease certificate issuance.
- To ensure lease certificates remain non-transferable and bound to tenant wallets.
- To gain hands-on experience with blockchain development and testing.

**Learnings:**

- I gained practical understanding of writing and deploying smart contracts using Solidity.
- I learned how ERC-721 NFTs can be customized for non-transferable certification use cases.
- I understood the importance of testing smart contracts before deployment to a blockchain network.

**Remark & Signature of Guide:**



Month No.: 4 (November 2025)

**Activities Performed:**

- I worked on developing the frontend interface of the AtLease project using React to enable user interaction with the system.
- I implemented screens for warehouse listing, lease request initiation, and certificate viewing.
- I integrated wallet connectivity to allow users to connect their blockchain wallets and view issued lease certificates.
- I began backend development using Node.js and Express.js by creating basic REST API endpoints.

**Observations:**

- To develop a functional user interface for interacting with the AtLease system.
- To establish backend APIs for communication between frontend and blockchain services.
- To ensure smooth interaction between user actions and system responses.

**Learnings:**

- I improved my understanding of building dynamic user interfaces using React.
- I learned how frontend and backend components communicate using REST APIs.
- I gained experience in handling wallet connectivity and blockchain interactions from the frontend.

**Remark & Signature of Guide:**

Month No.: 5 (December 2025)

**Activities Performed:**

- I continued backend development for the AtLease project by implementing core server-side logic using Node.js and Express.js.
- I developed REST API endpoints for handling lease certificate data and user requests.
- I integrated IPFS into the backend to store lease certificate metadata in a decentralized manner.
- I tested backend APIs using tools such as Postman and validated request and response handling.

**Observations:**

- To implement core backend functionality required for lease certificate issuance.
- To integrate decentralized storage and blockchain interaction with the backend.
- To ensure reliable communication between frontend, backend, and blockchain layers.

**Learnings:**

- I gained practical experience in developing REST APIs using Node.js and Express.js.
- I learned how to integrate IPFS and blockchain services into a backend application.
- I understood the importance of proper API testing and error handling.

**Remark & Signature of Guide:**

Month No.: 6 (January 2026)

**Activities Performed:**

- I continued backend development by enhancing existing API endpoints and improving overall application logic.
- I worked on integrating frontend components with backend services to ensure smooth data flow across the system.
- I implemented validation and error-handling mechanisms to improve reliability and security.
- I performed partial system testing to verify backend functionality and blockchain interactions.
- I reviewed the overall system and identified pending tasks related to deployment and optimization.

**Observations:**

- To complete remaining backend functionalities and improve system stability.
- To integrate all project components into a single working system.
- To prepare the application for final testing and deployment.

**Learnings:**

- I gained a better understanding of system integration and debugging in full-stack applications.
- I learned how to handle errors and edge cases in backend development.
- I understood the importance of testing and refinement before final deployment.

**Remark & Signature of Guide:**

Month No.: 7 (February 2026)

**Activities Performed:**

- I refined existing backend modules to improve performance, code structure, and maintainability.
- I optimized API endpoints and database queries to enhance system efficiency.
- I worked on improving integration between frontend and backend components for smoother communication.
- I implemented additional validation checks and improved error-handling mechanisms.
- I conducted system testing to identify bugs and inconsistencies in the application.

**Observations:**

- Refining existing modules significantly improved application performance and responsiveness.
- Continuous testing and debugging were essential for identifying hidden issues in the system.

**Learnings:**

- I learned the importance of refining and optimizing existing code for better system performance.
- I gained deeper understanding of debugging techniques and identifying performance bottlenecks.

**Remark & Signature of Guide:**

Month No.: 8 (March 2026)

**Activities Performed:**

- I designed the vendor login module to provide secure access for vendors within the AtLease system.
- I developed the frontend interface for vendor authentication using React and implemented form validation.
- I created backend API endpoints for vendor login and credential verification using Node.js and Express.js.
- I integrated the frontend with backend services to enable secure communication during the login process.
- I tested the login workflow and resolved issues related to authentication and error handling.

**Observations:**

- The vendor login module improved access control by allowing only authorized users to access vendor-specific features.
- Proper validation on both frontend and backend helped prevent invalid login attempts and improved system reliability.

**Learnings:**

- I gained practical experience in implementing authentication features in a full-stack application.
- I learned how frontend forms interact with backend APIs during the login process.

**Remark & Signature of Guide:**

Month No.: 9 (April 2026)

**Activities Performed:**

- We designed and developed the customer login module to provide secure access for customers within the AtLease system.
- We created the frontend interface for customer authentication using React and implemented input validation for login forms.
- We developed backend API endpoints for customer login and user credential verification using Node.js and Express.js.
- We integrated the customer login frontend with backend services to enable secure authentication and data exchange.
- We tested the complete login workflow and resolved issues related to validation, authentication, and API responses.

**Observations:**

- The customer login module enhanced system security by ensuring that only authorized customers could access their accounts.
- Proper integration between frontend and backend improved the reliability and responsiveness of the authentication process.

**Learnings:**

- We gained practical experience in implementing user authentication features in a full-stack application.
- We learned how to securely validate user credentials through backend APIs.
- We improved our understanding of frontend-backend integration and authentication workflows.

**Remark & Signature of Guide:**

Month No.: 10 (May 2026)

**Activities Performed:**

- We designed and developed the marketplace module for displaying available warehouses and open spaces for rent.
- We created frontend pages to showcase property details, including location, capacity, pricing, and availability information.
- We developed backend APIs to fetch and manage warehouse and open-space listings from the database.
- We integrated the frontend with backend services to display real-time property information to users.
- We tested the marketplace functionality and improved the user interface to provide a better browsing experience.
- related to validation, authentication, and API responses.

**Observations:**

- The marketplace module made it easier to organize and display available rental properties in a structured manner.
- Proper integration between frontend and backend ensured accurate and up-to-date listing information for users.

**Learnings:**

- We gained experience in developing marketplace-style interfaces for property listings.
- We learned how to efficiently retrieve and display large amounts of data through backend APIs.
- We improved our understanding of designing user-friendly interfaces for browsing and searching rental properties.

**Remark & Signature of Guide:**